The Sand-Reckoner and Shorter Works

ΑΡΧΙΜΗΔΟΥΣ ΨΑΜΜΙΤΗΣ · ΚΥΚΛΟΥ ΜΕΤΡΗΣΙΣ · ΠΡΟΒΛΗΜΑ ΒΟΕΙΚΟΝ

Archimedes

translated by Claude AI

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The Sand-Reckoner and Shorter Works

The Sand-Reckoner

I

S.1

Some people think, King Gelon, that the number of the sand is infinite in multitude: I mean not only the sand around Syracuse and the rest of Sicily, but also the sand in every land, inhabited and uninhabited alike. And there are others who do not take it to be infinite, yet hold that no number has been named so great as to exceed its multitude. It is clear that if those who are of this opinion were to conceive a volume of sand as large as the volume of the earth, with all the seas in it and the hollows of the earth filled up to a height equal to the highest of the mountains, they would be many times further still from supposing that any number could be stated which exceeds the multitude of that sand.

S.1

But I shall try to show you, by geometrical demonstrations which you will follow, that among the numbers named by us and published in the writings addressed to Zeuxippus there are some that exceed not only the number of the sand having a magnitude equal to the earth filled up as we have said, but also the number of the sand having a magnitude equal to the cosmos.

S.1

You know that most astronomers call by the name of cosmos the sphere whose centre is the centre of the earth and whose radius is equal to the straight line between the centre of the sun and the centre of the earth; for this is what you have heard in the proofs written by the astronomers.

S.1

But Aristarchus of Samos has published writings containing certain hypotheses, in which it follows from what is assumed that the cosmos is many times greater than the one just described. For he supposes that the fixed stars and the sun stay motionless; that the earth is carried round the sun along the circumference of a circle, the sun lying in the middle of its course; and that the sphere of the fixed stars, set about the same centre as the sun, is of such a size that the circle along which he supposes the earth to be carried round has to the distance of the fixed stars the same proportion as the centre of the sphere has to its surface.

S.1

Now this is plainly impossible: since the centre of the sphere has no magnitude, it cannot be thought to have any ratio to the surface of the sphere either. Aristarchus must be understood to mean this: since we take the earth to be, so to speak, the centre of the cosmos, the sphere containing the circle along which he supposes the earth to be carried round has to the sphere of the fixed stars the same ratio as the earth has to the cosmos as we have described it. For he fits the demonstrations of the phenomena to an assumption of this kind, and above all he appears to suppose that the sphere in which he makes the earth move is equal in magnitude to the cosmos as we have described it.

S.1

We say, then, that even if a sphere were made of sand, of the size that Aristarchus supposes the sphere of the fixed stars to have, even so it will be shown that some of the numbers in the Principles[1], those that have been given names, exceed in multitude the number of the sand having a magnitude equal to that sphere, when the following are assumed.

S.1

First, that the circumference of the earth is about 300 myriads of stadia and not greater, although some have tried to demonstrate, as you too are aware, that it is about 30 myriads of stadia. But I, going

beyond this and setting the size of the earth at about 10 times what earlier writers have held, assume its circumference to be about 300 myriads of stadia[2] and not greater.

S.1

After this, that the diameter of the earth is greater than the diameter of the moon, and the diameter of the sun greater than the diameter of the earth, taking in this the same view as most of the earlier astronomers.

S.1

After these, that the diameter of the sun is about 30 times the diameter of the moon and not greater, although among the earlier astronomers Eudoxus made it out to be about 9 times as great and Phidias my father[3] about 12 times, while Aristarchus tried to show that the diameter of the sun is more than 18 times, but less than 20 times, the diameter of the moon. But I, going beyond him too, so that what is proposed may be shown beyond dispute, assume the diameter of the sun to be about 30 times the diameter of the moon and not greater.

S.1

In addition to these, that the diameter of the sun is greater than the side of the 1,000-sided polygon inscribed in the greatest circle of those in the cosmos. This I assume because Aristarchus found the sun to appear as about 1/720 of the circle of the zodiac, while I myself, investigating in the following way, tried to obtain by means of instruments the angle into which the sun fits and which has its vertex at the eye.

S.1

Now to obtain the exact value is not easy, because neither the eye nor the hands nor the instruments by which it has to be obtained are trustworthy enough to declare the exact value. But on these matters it is not the moment to speak at length at present, especially as such things have often been pointed out. It is enough for me, for the demonstration of what is proposed, to obtain an angle that is not

greater than the angle into which the sun fits and which has its vertex at the eye, and again to obtain another angle that is not less than the angle into which the sun fits and which has its vertex at the eye.

S.1

So a long ruler was placed on an upright stand set in a place from which the sun would be seen as it rose, and a small cylinder was turned on the lathe and set upright on the ruler immediately after the rising of the sun. Then, while the sun was close to the horizon and could be looked at directly, the ruler was turned towards the sun and the eye was placed at the end of the ruler; and the cylinder, lying between the sun and the eye, hid the sun. Then, as it was moved away from the eye, the cylinder was fixed at the point where a little of the sun began to show on either side of it.

S.1

Now if it were the case that the eye saw from a single point, then, when straight lines were drawn from the end of the ruler, at the place where the eye was set, touching the cylinder, the angle contained by the lines drawn would be less than the angle into which the sun fits and which has its vertex at the eye, because something of the sun was seen past the cylinder on either side of it. But since the eyes do not see from a single point but from a certain magnitude, a rounded magnitude not less than the eye was taken, and, the magnitude being placed at the end of the ruler, at the place where the eye was set, straight lines were drawn touching both the magnitude and the cylinder; the angle contained by the lines drawn was therefore less than the angle into which the sun fits and which has its vertex at the eye.

S.1

The magnitude not less than the eye is found in the following way. Two thin little cylinders are taken, equal in thickness to one another, one white and the other not, and they are held in front of the eye, the white one at a distance from it, and the one that is not white as near to the eye as possible, so that it actually touches the face. If, then, the

little cylinders taken are thinner than the eye, the near cylinder is encompassed by the eye and the white one is seen by it: if they are much thinner, all of it; and if not much thinner, certain parts of the white one are seen on either side of the one near the eye. But when these little cylinders are taken of a suitable thickness, one of them hides the other and no more space than that. A magnitude, then, as great as the thickness of the cylinders that do this is, as nearly as may be, not less than the eye.

S.1

The angle not less than the angle into which the sun fits and which has its vertex at the eye was obtained as follows. The cylinder was set on the ruler at such a distance from the eye that the cylinder hid the whole of the sun, and straight lines were drawn from the end of the ruler, at the place where the eye was set, touching the cylinder; the angle contained by the straight lines drawn is then not less than the angle into which the sun fits and which has its vertex at the eye.

S.1

When a right angle was measured out by the angles obtained in this way, the larger[4] proved to be less than one part of the right angle divided into 164 parts, and the smaller greater than one part of the right angle divided into 200 parts. It is clear, then, that the angle into which the sun fits and which has its vertex at the eye is also less than one part of the right angle divided into 164, but greater than one part of the right angle divided into 200.

S.1

With these things accepted, it is shown that the diameter of the sun is greater than the side of the 1,000-sided polygon inscribed in the greatest circle of those in the cosmos.

S.1

For let a plane be conceived, drawn through the centre of the sun and the centre of the earth and through the eye, the sun being a little above the horizon; and let the plane so drawn cut the cosmos in the circle ΑΒΓ, the earth in the circle ΔΕΖ, and the sun in the circle ΣΗ; let

the centre of the earth be Θ and that of the sun K, and let the eye be Δ. Let straight lines be drawn touching the circle ΣH: from Δ the lines ΔΛ, ΔΞ, touching it at N and T, and from Θ the lines ΘM, ΘO, touching it at X and P; and let ΘM, ΘO cut the circle ABΓ at A and B.

S.1

Then ΘK is greater than ΔK, since the sun is assumed to be above the horizon; so that the angle contained by ΔΛ, ΔΞ is greater than the angle contained by ΘM, ΘO. But the angle contained by ΔΛ, ΔΞ is greater than 1/200 of a right angle, and less than one part of the right angle divided into 164; for it is equal to the angle into which the sun fits and which has its vertex at the eye. So the angle contained by ΘM, ΘO is less than one part of the right angle divided into 164, and the straight line AB is less than the line subtending one part of the circumference of the circle ABΓ when it is divided into 656 parts.

S.1

Now the perimeter of the said polygon has to the radius of the circle ABΓ a lesser ratio than 44 to 7, because the perimeter of every polygon inscribed in a circle has to the radius a lesser ratio than 44 to 7; for you know that it has been shown by us that the circumference of every circle is greater than 3 times the diameter by less than 1/7 of the diameter, and the perimeter of the inscribed polygon is less than this circumference. Therefore BA has to ΘK a lesser ratio than 11 to 1,148[5]; so that BA is less than 1/100 of ΘK.

S.1

But the diameter of the circle ΣH is equal to BA, because the half of BA, namely ΦA[6], is equal to KP; for, ΘK and ΘA being equal, perpendiculars have been drawn from their ends subtending the same angle. It is clear, then, that the diameter of the circle ΣH is less than 1/100 of ΘK. And the diameter EΘY is less than the diameter of the circle ΣH, since the circle ΔEZ is less than the circle ΣH; therefore ΘY and KΣ together are less than 1/100 of ΘK; so that ΘK has to YΣ a lesser ratio than 100 to 99[7].

S.1

And since ΘK is not less than ΘP, and ΣY is less than ΔT, ΘP would therefore have to ΔT a lesser ratio than 100 to 99. And since, the triangles ΘKP, ΔKT being right-angled, the sides KP, KT are equal while ΘP, ΔT are unequal and ΘP is the greater, the angle contained by ΔT, ΔK has to the angle contained by ΘP, ΘK a greater ratio than ΘK has to ΔK, but a lesser ratio than ΘP has to ΔT. For if, in two right-angled triangles, one pair of the sides about the right angle are equal and the other pair unequal, the greater of the angles at the unequal sides has to the lesser a greater ratio than the greater of the lines subtending the right angle has to the lesser, but a lesser ratio than the greater of the lines about the right angle has to the lesser.

S.1

So the angle contained by ΔΛ, ΔΞ has to the angle contained by ΘO, ΘM a lesser ratio than ΘP has to ΔT, which itself has a lesser ratio than 100 to 99; so that the angle contained by ΔΛ, ΔΞ also has to the angle contained by ΘM, ΘO a lesser ratio than 100 to 99. And since the angle contained by ΔΛ, ΔΞ is greater than 1/200 of a right angle, the angle contained by ΘM, ΘO would be greater than 99 parts of the right angle divided into 20,000; so that it is greater than one part of the right angle divided into 203. Therefore BA is greater than the line subtending one part of the circumference of the circle ABΓ when it is divided into 812 parts. But the diameter of the sun is equal to AB; it is clear, then, that the diameter of the sun is greater than the side of the 1,000-sided polygon.

II

S.2

With these things assumed, the following too are shown: namely that the diameter of the cosmos is less than 10,000 times the diameter of the earth, and further that the diameter of the cosmos is less than 100 myriad myriads of stadia.

S.2

For since it is assumed that the diameter of the sun is not greater than 30 times the diameter of the moon, and that the diameter of the earth is greater than the diameter of the moon, it is clear that the diameter of the sun is less than 30 times the diameter of the earth. Again, since the diameter of the sun was shown to be greater than the side of the 1,000-sided polygon inscribed in the greatest circle of those in the cosmos, it is evident that the perimeter of the said 1,000-sided polygon is less than 1,000 times the diameter of the sun. But the diameter of the sun is less than 30 times the diameter of the earth; so that the perimeter of the 1,000-sided polygon is less than 30,000 times the diameter of the earth.

S.2

Since, then, the perimeter of the 1,000-sided polygon is less than 30,000 times the diameter of the earth, but greater than 3 times the diameter of the cosmos (for it has been shown to you that the diameter of every circle is less than 1/3 of the perimeter of every polygon inscribed in the circle that is equilateral and has more angles than the hexagon), the diameter of the cosmos would be less than 10,000 times the diameter of the earth.

S.2

That the diameter of the cosmos is less than 10,000 times the diameter of the earth has thus been shown; and that the diameter of the cosmos is less than 100 myriad myriads of stadia is clear from the following. Since it is assumed that the circumference of the earth is not greater than 300 myriads of stadia, and the circumference of the earth is greater than 3 times its diameter, because the circumference of every circle is greater than 3 times the diameter, it is clear that the diameter of the earth is less than 100 myriads of stadia[8]. Since, then, the diameter of the cosmos is less than 10,000 times the diameter of the earth, it is clear that the diameter of the cosmos is less than 100 myriad myriads of stadia.

S.2

About the magnitudes and the distances, then, these are my assumptions; and about the sand, the following: that if there is a magnitude made up of sand not greater than a poppy-seed, the number of the sand in it is not greater than 10,000, and that the diameter of the poppy-seed is not less than 1/40 of a finger-breadth.

S.2

I assume this after investigating in the following way. Poppy-seeds were placed on a smooth ruler in a straight line, lying in a single row and touching one another, and 25 poppy-seeds took up more room than the length of one finger-breadth. Making the diameter of the poppy-seed smaller, then, I assume it to be about 1/40 of a finger-breadth and not less, wishing in this way too to have what is proposed shown as far beyond dispute as possible.

III

S.3

These, then, are the things I assume. But I take it to be useful for the naming of the numbers to be stated, so that other readers too, who have not come upon the book addressed to Zeuxippus, may not go astray because nothing has been said about it beforehand in this book.

S.3

Now it happens that the names of the numbers have been handed down to us as far as a myriad, and beyond a myriad we recognise them well enough, speaking of a number of myriads up to a myriad myriads. Let the numbers just mentioned, then, up to a myriad myriads, be called by us first numbers; and let a myriad myriads of the first numbers be called the unit of the second numbers, and let there be counted of the second numbers units, and from the units tens and hundreds and thousands and myriads, up to a myriad myriads. Again, let a myriad myriads of the second numbers be called the unit of the third numbers, and let there be counted of the third numbers units, and from the units tens and hundreds and thousands and myriads, up to a myriad myriads. In the same way let a myriad myriads of the

third numbers be called the unit of the fourth numbers, and a myriad myriads of the fourth numbers be called the unit of the fifth numbers, and let the numbers, advancing always in this way, have their names up to a myriad myriads of the myriad-myriadth numbers.

S.3

The numbers known thus far are indeed sufficient, but it is possible to go on further still. For let the numbers just mentioned be called numbers of the first period, and let the last number of the first period be called the unit of the first numbers of the second period. Again, let a myriad myriads of the first numbers of the second period be called the unit of the second numbers of the second period. Likewise let the last of these be called the unit of the third numbers of the second period, and let the numbers, advancing always in this way, have their names in the second period up to a myriad myriads of the myriad-myriadth numbers. Again, let the last number of the second period be called the unit of the first numbers of the third period, and let them advance always in this way up to a myriad myriads of the myriad-myriadth numbers of the myriad-myriadth period.

S.3

With the numbers named in this way, if there are numbers set out in order in proportion starting from the unit, and the one next to the unit is 10, the first 8 of them, counting the unit, will belong to the numbers called first, and the next 8 after them to those called second, and the rest, in the same way as these, will belong to the numbers called by the name that matches the distance of their octad of numbers from the first octad of numbers.

S.3

Now the 8th number of the first octad of numbers is 1,000 myriads, and the first of the second octad, since it is 10 times the one before it, will be a myriad myriads; and this is the unit of the second numbers. The 8th of the second octad is 1,000 myriads of the second numbers. Again, the first of the third octad, since it is 10 times the one before it, will be a myriad myriads of the second numbers; and this is the unit of

the third numbers. And it is evident that the octads, however many there are, will be as has been said.

S.3

It is useful also to know the following. If, of numbers that are in proportion starting from the unit, some from the same proportion multiply one another, the product will be from the same proportion, distant from the greater of the numbers that multiplied one another by as many terms as the lesser of the numbers that multiplied is distant in the proportion from the unit; and it will be distant from the unit by one term fewer than the number, taken both together, of the terms by which the numbers that multiplied one another are distant from the unit.

S.3

For let there be some numbers in proportion starting from the unit, Α, Β, Γ, Δ, Ε, Ζ, Η, Θ, Ι, Κ, Λ, and let Α be the unit; and let Δ be multiplied by Θ, and let the product be Χ. Let Λ, then, be taken from the proportion, distant from Θ by as many terms as Δ is distant from the unit. It is to be shown that Χ is equal to Λ.

S.3

Since, then, the numbers being in proportion, Δ is distant from Α by as many as Λ is from Θ, Δ has to Α the same ratio as Λ has to Θ. But Δ is Α multiplied by Δ; therefore Λ also is Θ multiplied by Δ; so that Λ is equal to Χ. It is clear, then, that the product both is from the proportion and is distant from the greater of the numbers that multiplied one another by as many terms as the lesser is distant from the unit.

S.3

And it is evident that it is also distant from the unit by one term fewer than the number, taken both together, of the terms by which Δ and Θ are distant from the unit. For Α, Β, Γ, Δ, Ε, Ζ, Η, Θ are as many as Θ is distant from the unit, and Ι, Κ, Λ are one fewer than Δ is distant from the unit; for with Θ they are as many as that.

IV

S.4

These things being in part assumed and in part demonstrated, what was proposed will be shown.

S.4

For since it is assumed that the diameter of the poppy-seed is not less than 1/40 of a finger-breadth, it is clear that the sphere having a diameter of one finger-breadth is no greater than will hold 64,000 poppy-seeds; for it is a multiple, by the number just stated, of the sphere having a diameter of 1/40 of a finger-breadth; for it has been shown to you that spheres have to one another the triplicate ratio of their diameters. And since it is also assumed that the number of the sand that goes into the magnitude of the poppy-seed is not greater than 10,000, it is clear that, if the sphere having a diameter of one finger-breadth were filled with sand, the number of the sand would not be greater than 10,000 times 64,000. And this number is 6 units of the second numbers[9] and 4,000 myriads of the first; it is therefore less than 10 units of the second numbers.

S.4

The sphere having a diameter of 100 finger-breadths is a multiple of the sphere having a diameter of one finger-breadth by 100 myriads, because spheres have to one another the triplicate ratio of their diameters. If, then, a sphere were made of sand as great in size as the sphere having a diameter of 100 finger-breadths, it is clear that the number of the sand will be less than the number produced when the 10 units of the second numbers are multiplied by the 100 myriads. And since the 10 units of the second numbers are the 10th number from the unit in proportion, in the proportion whose terms are each 10 times the one before, and the 100 myriads are the 7th from the unit in the same proportion, it is clear that the number produced will be the 16th from the unit of those in the same proportion; for it has been shown that it is distant from the unit by one term fewer than the number, taken both together, of the terms by which the numbers that

multiplied one another are distant from the unit. Of these 16, the first 8, counting the unit, belong to the numbers called first, and the 8 after them to the second, and the last of them is 1,000 myriads of the second numbers. It is evident, then, that the multitude of the sand having a magnitude equal to the sphere whose diameter is 100 finger-breadths is less than 1,000 myriads of the second numbers.

S.4

Again, the sphere having a diameter of 10,000 finger-breadths is a multiple of the sphere having a diameter of 100 finger-breadths by 100 myriads. If, then, a sphere were made of sand as great in size as the sphere having a diameter of 10,000 finger-breadths, it is clear that the number of the sand will be less than the number produced when the 1,000 myriads of the second numbers are multiplied by the 100 myriads. And since the 1,000 myriads of the second numbers are the 16th number from the unit in proportion, and the 100 myriads are the 7th from the unit in the same proportion, it is clear that the number produced will be the 22nd from the unit of those in the same proportion. Of these 22, the first 8, counting the unit, belong to the numbers called first, the 8 after them to those called second, and the remaining 6 to those called third, and the last of them is 10 myriads of the third numbers. It is evident, then, that the multitude of the sand having a magnitude equal to the sphere whose diameter is 10,000 finger-breadths is less than 10 myriads of the third numbers.

S.4

And since the sphere having a diameter of one stadium is less than the sphere having a diameter of 10,000 finger-breadths, it is clear that the multitude of the sand having a magnitude equal to the sphere whose diameter is one stadium is also less than 10 myriads of the third numbers.

S.4

Again, the sphere having a diameter of 100 stadia is a multiple of the sphere having a diameter of one stadium by 100 myriads. If, then, a sphere were made of sand as great in size as the sphere having a

diameter of 100 stadia[10], it is clear that the number of the sand will be less than the number produced when the 10 myriads of the third numbers are multiplied by the 100 myriads. And since the 10 myriads of the third numbers are the 22nd from the unit in proportion, and the 100 myriads are the 7th from the unit in the same proportion, it is clear that the number produced will be the 28th from the unit in the same proportion. Of these 28, the first 8, counting the unit, belong to the numbers called first, the next 8 after them to the second, the 8 after those to the third, and the remaining 4 to those called fourth, and the last of them is 1,000 units of the fourth numbers. It is evident, then, that the multitude of the sand having a magnitude equal to the sphere whose diameter is 100 stadia is less than 1,000 units of the fourth numbers.

S.4

Again, the sphere having a diameter of 10,000 stadia is a multiple of the sphere having a diameter of 100 stadia by 100 myriads. If, then, a sphere were made of sand as great in size as the sphere having a diameter of 10,000 stadia, it is clear that the multitude of the sand will be less than the number produced when the 1,000 units of the fourth numbers are multiplied by the 100 myriads. And since the 1,000 units of the fourth numbers are the 28th from the unit in proportion, and the 100 myriads are the 7th from the unit in the same proportion, it is clear that the number produced will be the 34th from the unit in the same proportion. Of these 34, the first 8, counting the unit, belong to the numbers called first, the 8 after them to the second, the next 8 after those to the third, the 8 after those to the fourth, and the remaining 2 will belong to those called fifth, and the last of them is 10 units of the fifth numbers. It is clear, then, that the multitude of the sand having a magnitude equal to the sphere whose diameter is 10,000 stadia will be less than 10 units of the fifth numbers.

S.4

Again, the sphere having a diameter of 100 myriads of stadia is a multiple of the sphere having a diameter of 10,000 stadia by 100

myriads. If, then, a sphere were made of sand as great in size as the sphere having a diameter of 100 myriads of stadia, it is clear that the number of the sand will be less than the number produced when the 10 units of the fifth numbers are multiplied by the 100 myriads. And since the 10 units of the fifth numbers are the 34th from the unit in proportion, and the 100 myriads are the 7th from the unit in the same proportion, it is clear that the number produced will be the 40th from the unit in the same proportion. Of these 40, the first 8, counting the unit, belong to the numbers called first, the next 8 after them to the second, the next 8 after those to the third, the 8 after the third to the fourth, and the 8 after those to the numbers called fifth, and the last of them is 1,000 myriads of the fifth numbers. It is evident, then, that the multitude of the sand having a magnitude equal to the sphere whose diameter is 100 myriads of stadia is less than 1,000 myriads of the fifth numbers.

S.4

The sphere having a diameter of a myriad myriads of stadia is a multiple of the sphere having a diameter of 100 myriads of stadia by 100 myriads. If, then, a sphere were made of sand as great in size as the sphere having a diameter of a myriad myriads of stadia, it is evident that the multitude of the sand will be less than the number produced when the 1,000 myriads of the fifth numbers are multiplied by the 100 myriads. And since the 1,000 myriads of the fifth numbers are the 40th from the unit in proportion, and the 100 myriads are the 7th from the unit in the same proportion, it is clear that the number produced will be the 46th from the unit. Of these 46, the first 8, counting the unit, belong to the numbers called first, the 8 after them to the second, the next 8 after those to the third, the next 8 after the third to the fourth, the 8 after the fourth to the fifth, and the remaining 6 belong to those called sixth, and the last of them is 10 myriads of the sixth numbers. It is evident, then, that the multitude of the sand having a magnitude equal to the sphere whose diameter is a myriad myriads of stadia is less than 10 myriads of the sixth numbers.

S.4

The sphere having a diameter of 100 myriad myriads of stadia is a multiple of the sphere having a diameter of a myriad myriads of stadia by 100 myriads. If, then, a sphere were made of sand as great in size as the sphere having a diameter of 100 myriad myriads of stadia, it is evident that the multitude of the sand will be less than the number produced when the 10 myriads of the sixth numbers are multiplied by the 100 myriads. And since the 10 myriads of the sixth numbers are the 46th from the unit in proportion, and the 100 myriads are the 7th from the unit in the same proportion, it is clear that the number produced will be the 52nd from the unit in the same proportion. Of these 52, the first 48, counting the unit, are the numbers called first and second and third and fourth and fifth and sixth, and the remaining 4 belong to those called seventh, and the last of them is 1,000 units of the seventh numbers. It is evident, then, that the multitude of the sand having a magnitude equal to the sphere whose diameter is 100 myriad myriads of stadia is less than 1,000 units of the seventh numbers.

S.4

Since, then, the diameter of the cosmos was shown to be less than 100 myriad myriads of stadia, it is clear that the multitude of the sand having a magnitude equal to the cosmos is also less than 1,000 units of the seventh numbers.

S.4

That the multitude of the sand having a magnitude equal to what most astronomers call the cosmos is less than 1,000 units of the seventh numbers has thus been shown; and that the multitude of the sand having a magnitude equal to a sphere of the size that Aristarchus supposes the sphere of the fixed stars to have is less than 1,000 myriads of the eighth numbers will be shown next.

S.4

For since it is assumed that the earth has to the cosmos as described by us the same ratio as the said cosmos has to the sphere of the fixed

stars which Aristarchus supposes, the diameters of the spheres also have the same ratio to one another. But the diameter of the cosmos has been shown to be less than 10,000 times the diameter of the earth; it is clear, then, that the diameter of the sphere of the fixed stars is also less than 10,000 times the diameter of the cosmos. And since spheres have to one another the triplicate ratio of their diameters, it is evident that the sphere of the fixed stars which Aristarchus supposes is a multiple of the cosmos by less than a myriad times a myriad myriads.

S.4

But it has been shown that the multitude of the sand having a magnitude equal to the cosmos is less than 1,000 units of the seventh numbers; it is clear, then, that if a sphere were made of sand, of the size that Aristarchus supposes the sphere of the fixed stars to have, the number of the sand will be less than the number produced when the 1,000 units are multiplied by a myriad times a myriad myriads. And since the 1,000 units of the seventh numbers are the 52nd from the unit in proportion, and a myriad times a myriad myriads is the 13th from the unit in the same proportion, it is clear that the number produced will be the 64th from the unit in the same proportion; and this is the 8th of the eighth numbers, which would be 1,000 myriads of the eighth numbers. It is evident, therefore, that the multitude of the sand having a magnitude equal to the sphere of the fixed stars which Aristarchus supposes is less than 1,000 myriads of the eighth numbers.

S.4

These things, King Gelon, I suppose will seem hard to believe to the many who have had no share in mathematics, but to those who have taken part in it and have given thought to the distances and the sizes of the earth and the sun and the moon and the whole cosmos they will be credible because of the demonstration; and that is why I thought it not unfitting for you too to consider them.

Measurement of a Circle

Proposition 1

C.1

Every circle is equal to a right-angled triangle in which the radius is equal to one of the sides about the right angle and the circumference to the base.

C.1

Let the circle ΑΒΓΔ stand to the triangle Ε as is supposed; I say that it is equal to it.

C.1

For, if possible, let the circle be greater, and let the square ΑΓ be inscribed, and let the arcs be bisected, and let the segments be by now less than the excess by which the circle exceeds the triangle; the rectilinear figure is therefore still greater than the triangle.

C.1

Let the centre Ν be taken, and the perpendicular ΝΞ[11]; then ΝΞ is less than the side of the triangle. And the perimeter of the rectilinear figure is also less than the remaining side, since it is less even than the circumference of the circle. Therefore the rectilinear figure is less than the triangle Ε; which is absurd.

C.1

Next let the circle, if possible, be less than the triangle Ε, and let the square be circumscribed, and let the arcs be bisected, and let tangents be drawn through the points; the angle ΟΑΡ[12] is therefore right. Therefore ΟΡ is greater than ΜΡ, for ΡΜ is equal to ΡΑ; and therefore the triangle ΡΟΠ is greater than half of the figure ΟΖΑΜ.

C.1

Let the sectors similar to the sector ΠΖΑ be left less than the excess by which Ε exceeds the circle ΑΒΓΔ; the circumscribed rectilinear figure is therefore still less than Ε; which is absurd. For it is greater, because

NA is equal to the perpendicular of the triangle, while the perimeter is greater than the base of the triangle.

C.1

Therefore the circle is equal to the triangle E.

Proposition 2

C.2

The circle has to the square on its diameter the ratio which 11 has to 14.

C.2

Let there be a circle whose diameter is AB, and let the square ΓH be circumscribed, and let ΔE be double ΓΔ, and EZ 1/7 of ΓΔ[13].

C.2

Since, then, the triangle AΓE has to the triangle AΓΔ the ratio which 21 has to 7, while AΓΔ has to AEZ the ratio which 7 has to 1, AΓZ is to AΓΔ as 22 to 7. But the square ΓH is 4 times AΓΔ, and the triangle AΓZ[14] is equal to the circle AB; therefore the circle has to the square ΓH the ratio which 11 has to 14.

Proposition 3

C.3

The circumference of every circle is 3 times the diameter and exceeds this further by less than 1/7 of the diameter, but by more than 10/71 of it.

C.3

Let there be a circle, and let AΓ be a diameter and E the centre, and ΓΛZ a tangent, and let the angle ZEΓ be 1/3 of a right angle. Then EZ has to ZΓ the ratio which 306 has to 153, and EΓ has to ΓZ a greater ratio than 265 to 153[15].

C.3

Now let the angle ZEΓ be bisected by EH. Then as ZE is to EΓ, so is ZH to HΓ. Therefore as ZE and EΓ together are to ZΓ, so is EΓ to ΓH; so that ΓE has to ΓH a greater ratio than 571 to 153. Therefore EH has to HΓ in

square the ratio which 349,450 has to 23,409[16]; in length, therefore, that which 591 1/8 has to 153.

C.3

Again, let the angle ΗΕΓ be bisected by ΕΘ. For the same reasons, then, ΕΓ has to ΓΘ a greater ratio than 1,162 1/8 to 153; therefore ΘΕ has to ΘΓ a greater ratio than 1,172 1/8 to 153.

C.3

Further, let the angle ΘΕΓ be bisected by ΕΚ. Then ΕΓ has to ΓΚ a greater ratio than 2,334 1/4 to 153; therefore ΕΚ has to ΓΚ a greater ratio than 2,339 1/4 to 153.

C.3

Further, let the angle ΚΕΓ be bisected by ΛΕ. Then ΕΓ has to ΛΓ a greater ratio than 4,673 1/2 to 153.

C.3

Since, then, the angle ΖΕΓ, being 1/3 of a right angle, has been bisected 4 times, the angle ΛΕΓ is 1/48 of a right angle. Let the angle ΓΕΜ, then, be set equal to it at Ε; the angle ΛΕΜ is therefore 1/24 of a right angle. And therefore the straight line ΛΜ is a side of the polygon of 96 sides about the circle.

C.3

Since, then, ΕΓ was shown to have to ΓΛ a greater ratio than 4,673 1/2 to 153, while ΑΓ is double ΕΓ and ΛΜ double ΓΛ, ΑΓ also therefore has to the perimeter of the 96-sided polygon a greater ratio than 4,673 1/2 to 14,688. And the latter is 3 times the former with 667 1/2 over, which is less than 1/7 of 4,673 1/2; so that the polygon about the circle is 3 times the diameter and greater by less than 1/7 of it. All the more, therefore, is the circumference of the circle less than 3 times the diameter and 1/7 of it over.

C.3

Let there be a circle, and let ΑΓ be a diameter, and the angle ΒΑΓ 1/3 of a right angle. Then ΑΒ has to ΒΓ a lesser ratio than 1,351 to 780, and ΑΓ has to ΓΒ the ratio which 1,560 has to 780[17].

C.3

Let the angle ΒΑΓ be bisected by ΑΗ. Since, then, the angle ΒΑΗ is equal to the angle ΗΓΒ, but also to the angle ΗΑΓ, the angle ΗΓΒ is also equal to the angle ΗΑΓ. And the right angle ΑΗΓ is common; therefore the third angle ΗΖΓ[18] is also equal to the third angle ΑΓΗ. Therefore the triangle ΑΗΓ is equiangular with the triangle ΓΗΖ; therefore, as ΑΗ is to ΗΓ, so is ΓΗ to ΗΖ and ΑΓ to ΓΖ.

C.3

But as ΑΓ is to ΓΖ, so are ΓΑ and ΑΒ together to ΒΓ; therefore also, as ΒΑ and ΑΓ together are to ΒΓ, so is ΑΗ to ΗΓ. For this reason, then, ΑΗ has to ΗΓ a lesser ratio than 2,911 to 780, and ΑΓ has to ΓΗ a lesser ratio than 3,013 3/4 to 780.

C.3

Let the angle ΓΑΗ be bisected by ΑΘ. Then, for the same reasons, ΑΘ has to ΘΓ a lesser ratio than 5,924 3/4 to 780, or than 1,823 to 240; for each of these is 4/13 of each of those. So that ΑΓ has to ΓΘ a lesser ratio than 1,838 9/11 to 240.

C.3

Further, let the angle ΘΑΓ be bisected by ΚΑ; and ΑΚ has to ΚΓ a lesser ratio than 1,007 to 66; for each of these is 11/40 of each of those. Therefore ΑΓ has to ΚΓ a lesser ratio than 1,009 1/6 to 66.

C.3

Further, let the angle ΚΑΓ be bisected by ΛΑ. Then ΑΛ has to ΛΓ a lesser ratio than 2,016 1/6 to 66, and ΑΓ has to ΓΛ a lesser ratio than 2,017 1/4 to 66.

C.3

Inversely, therefore, the perimeter of the polygon has to the diameter a greater ratio than 6,336 to 2,017 1/4, and 6,336 is greater than 3 times 2,017 1/4 and 10/71 of it besides. Therefore the perimeter of the 96-sided polygon in the circle is also 3 times the diameter and greater by more than 10/71 of it; so that the circle too is all the more 3 times the diameter and greater by more than 10/71 of it.

C.3

Therefore the circumference of the circle is 3 times the diameter and greater by less than 1/7 of it, but by more than 10/71 of it.

The Cattle Problem

B

A problem which Archimedes devised in the form of an epigram and sent, in his letter to Eratosthenes of Cyrene, to those in Alexandria who busy themselves with such matters, for them to work out.

B

Reckon up, stranger, the multitude of the cattle of the Sun, setting your mind to it, if you have a share of wisdom: how great a number once grazed on the plains of the Sicilian island of Thrinacia, divided into four herds of differing colour, one milk-white, another gleaming with a dark hue, a third tawny, and the fourth dappled.

B

In each herd there were bulls, abounding in numbers and standing in the following proportion. Understand, stranger, that the white-haired bulls were equal to a half and a third of the dark bulls together with all the tawny; while the dark were equal to the fourth part and the fifth of the dappled, with all the tawny besides. And observe that those remaining, the dappled, were equal to a sixth part and a seventh of the white bulls together with all the tawny.

B

With the cows it was thus. The white-haired were exactly equal to the third part and the fourth of the whole dark herd; while the dark cows in their turn were equal to the fourth part and the fifth together of the dappled, when all of these, bulls included, went to pasture[19]. The dappled cows, in the fourfold division[20], had a number equal to a fifth part and a sixth of the tawny-haired herd. And the tawny cows were counted equal to half of a third part of the white herd, and a seventh part.

B

Stranger, if you tell exactly how many the cattle of the Sun were, giving separately the number of the well-fed bulls, and separately again how many cows there were of each colour, you would be called neither unknowing nor unskilled in numbers; yet you would not so far be numbered among the wise.

B

But come, consider also all these further facts about the cattle of the Sun. When the white-haired bulls mingled their multitude with the dark, they stood firm, of equal measure in depth and in breadth, and the plains of Thrinacia, stretching far on every side, were filled with their block[21]. And again, when the tawny and the dappled were gathered into one, they stood rising rank upon rank, beginning from one, and completed the three-sided figure[22], with no bulls of other colours among them and none wanting.

B

When you have found all this out together and gathered it up in your mind, and have rendered, stranger, all the measures of the multitudes, go your way exulting as a victor, and know for certain that you have been judged rich indeed in this kind of wisdom.

Notes

[1] The Greek as printed has ἐν ἀρχᾷ, literally "in the beginning"; it is generally taken as a reference by title, Principles (Ἀρχαί), to the work addressed to Zeuxippus mentioned above and again in III.

[2] The numeral has dropped out of the Greek at this second statement; 300 is restored from the first statement just above and from the 300 myriads written out in words in II.

[3] "My father" renders an emendation (ἁμοῦ πατρός), attributed to Blass and generally accepted, of a manuscript reading that gives no sense (reported as Ἀκούπατρος).

[4] The Greek is corrupt at this point (ἁ ἐν στίγῳ as printed); "the larger" is the sense required by the answering "the smaller".

[5] The numeral is printed without its thousands-sign; 1,148 is confirmed by the reckoning, since 44 to 7 for the whole perimeter gives, for one side out of 656, 44 to 4,592, that is 11 to 1,148.

[6] Three points of the lost diagram are never set out in the text: Φ is the midpoint of AB, where ΘK crosses it, and Υ and Σ, used just below, are the points where ΘK meets the circle of the earth and the circle of the sun.

[7] The sign for 90 has been lost from this numeral in all five places where it occurs. That it is 99 is certain: ΘΥ and ΚΣ together are less than 1/100 of ΘΚ, and 99 parts in 20,000 are more than 1 part in 203.

[8] The numeral is missing in the Greek; 100 is required by the argument (a circumference of not more than 300 myriads is more than 3 times the diameter) and by the 100 myriad myriads that follow.

[9] The sign for 6 is lost in the Greek; it is certain, since 10,000 times 64,000 is 640,000,000, that is 6 units of the second numbers and 4,000 myriads of the first.

[10] The numeral 100 has dropped out of the Greek here and in the conclusion of this stage, and is restored from its opening; further on the sign for 1,000 is once printed as that for 1 in "1,000 units of the seventh numbers".

[11] N is the centre of the circle and NΞ the perpendicular from it to a side of the inscribed figure. "The side of the triangle" is the side equal to the radius, and "the remaining side" the one equal to the circumference.

[12] The lettering of the lost figure has to be inferred: O is a corner of the circumscribed square, Z and M the points where the two sides through O touch the circle, A the middle of the arc ZM, and Π, P the points where the tangent at A cuts those sides. The "sectors" below are the pieces left outside the circle, such as that bounded by ΠZ, ΠA and the arc ZA.

[13] The figure is not fully set out: ΓΔ is a side of the square, with Δ, E, Z following Γ on the same straight line, and A is the middle of the side that meets it at Γ, so that AΓ is equal to the radius.

[14] So the Greek; the triangle meant is AΓZ. It is equal to the circle only if the circumference is taken as 3 1/7 times the diameter, as a clause deleted by the editors and omitted here points out, so that this proposition depends on Proposition 3.

[15] Here, and at 591 1/8 to 153 below, the Greek states the ratio simply; "a greater ratio than" is to be understood, as the sequel shows.

[16] The multipliers of the myriad-sign have dropped out of the Greek here and in 14,688 below; the values are fixed by the reckoning (571 squared plus 153 squared, 153 squared, and 96 times 153). The sign for 90 is likewise damaged in 591 1/8 and in each mention of the 96-sided polygon, and 1,351 is printed with two letters transposed.

[17] This clause is bracketed by the editors but is kept here, since the 2,911 that follows is the sum of 1,351 and 1,560.

[18] Z is not set out in the text: it is the point where AH cuts BΓ. H, Θ, K, Λ lie on the circle, and the text passes over the step that ΓΛ is a side of the inscribed 96-sided polygon, whose perimeter is therefore 96 times 66, or 6,336.

[19] In each condition on the cows the fraction is taken of the whole herd of the colour named, bulls and cows together. In this line the Greek as printed (πάσαις ... ἐρχομέναις) does not construe with "the dappled" and has been emended, but the sense is not in doubt.

[20] The Greek word (τετραχῇ, "in four parts") is obscure here and may be corrupt.

[21] The Greek word (πλίνθου, "brick", that is a rectangular block) is doubtful. Whether the white and dark bulls together are to form a square number or only a rectangular one is disputed, and the size of the solution depends on it.

[22] That is, the tawny and dappled bulls together make a triangular number, the rows being of 1, 2, 3 and so on.